

Notes: De Nardi, French, and Jones (AER, 2016)

Medicaid Insurance in Old Age

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Data and measurement	3
3.1	AHEAD data	3
3.2	MCBS data	3
3.3	Medicaid reciprocity, payments, and taxes	4
3.4	Medical spending and assets	4
4	Model and empirical specifications	4
4.1	Preferences and medical needs	4
4.2	Insurance and Medicaid floors	5
4.3	Dynamic problem	5
4.4	Estimation	5
5	Identification and empirical design	6
5.1	What identifies Medicaid generosity and bequest motives?	6
5.2	What identifies risk aversion and medical demand?	6
5.3	Policy counterfactuals	6
6	Tables and figures	7
6.1	Figure 1 and Table 1: Medicaid reciprocity and benefits	7
6.2	Figures 2 and 3: Out-of-pocket medical spending risk	7
6.3	Figures 4 and 5: Assets and income profiles	8
6.4	Tables 2 and 3: Mortality and nursing-home risk	9
6.5	Table 4 and Figure 6: Estimated parameters and Medicaid model fit	10
6.6	Figures 7 and 8: Asset model fit	10
6.7	Figures 9 and 10: Medical spending model fit	11
6.8	Table 5 and Figure 11: Average spending and simulated consumption	12
6.9	Tables 6 and 7: Lifetime spending, transfers, and taxes	13

6.10	Tables 8 and 9: Medicaid cuts and expansions	13
6.11	Tables 10 and 11: Moral hazard and selection into Medicaid	14
7	Short summary	15
8	Conclusion	16
8.1	Core conclusion	16
8.2	Economic intuition	16
8.3	Incremental contribution revisited	16
8.4	Policy implications	16
8.5	Limitations	16
8.6	Takeaway	17

1 Big picture

- **Question.** How does Medicaid insure elderly single retirees against medical spending risk, who receives transfers, who pays taxes to finance them, and how much do retirees value this insurance?
- **Setting.** The paper focuses on currently single retirees. This group is central for Medicaid in old age because singles are more likely to enter nursing homes and to exhaust resources late in life.
- **Main empirical facts.** Low permanent-income retirees receive Medicaid frequently throughout retirement, while higher-income retirees rarely receive Medicaid early in old age but can enter Medicaid at advanced ages after catastrophic medical shocks. Conditional on receiving Medicaid, high-income recipients receive especially large transfers.
- **Core model.** The authors estimate a life-cycle model in which retirees choose saving, nonmedical consumption, medical spending, and whether to apply for Medicaid. Retirees face health, mortality, nursing home, and medical-need risks.
- **Medicaid design.** The paper explicitly models two pathways into Medicaid: the *categorically needy* route for the lifelong poor, including SSI, and the *medically needy* route for retirees impoverished by high medical needs.
- **Key valuation result.** Retirees value current Medicaid insurance at more than its actuarial cost, but most retirees would value an expansion of Medicaid at less than the additional cost.
- **Distributional result.** Medicaid is strongly redistributive ex ante, but it also provides valuable catastrophic insurance for richer retirees because rare late-life medical shocks are very costly in utility terms.
- **Policy takeaway.** For current single retirees, the paper concludes that Medicaid is roughly the right size: cutting it is costly in welfare terms, but expanding it is generally valued less than it costs.

2 Incremental contribution of the paper

- **From spending facts to insurance valuation.** Earlier work documents that late-life medical spending risk is large. This paper asks how Medicaid transfers and Medicaid insurance are valued over the remaining lifetime of old singles.
- **A richer Medicaid model.** The paper goes beyond a simple consumption floor by modeling Medicaid as a medical-needs-contingent utility floor and by distinguishing categorically needy and medically needy eligibility.
- **Endogenous medical spending.** Medical spending is a choice, not an exogenous shock. This lets the authors study both insurance and moral hazard: Medicaid changes incentives to save and to consume medical care.
- **New data combination.** The authors combine the AHEAD/HRS panel, the MCBS, and the PSID. The AHEAD provides panel data on assets, health, nursing homes, and out-of-pocket spending; the MCBS provides payer-specific medical spending; the PSID is used to allocate Medicaid-related taxes.

- **Identification contribution.** Matching Medicaid reciprocity rates introduces a new source of identification for the generosity of the Medicaid floor and for the bequest motive. A generous Medicaid floor reduces precautionary saving, so matching high assets requires a bequest motive.
- **Policy-counterfactual contribution.** The paper computes compensating variation for Medicaid cuts and expansions, compares valuations to actuarial costs, and then compares both to the tax burden implied by the financing system.
- **Selection contribution.** The model quantifies how Medicaid recipients differ from nonrecipients in income and medical needs, showing that recipients are much more likely to face extreme medical need shocks.

3 Data and measurement

3.1 AHEAD data

- The AHEAD sample is drawn from the Health and Retirement Study and follows people who were aged 70 or older and noninstitutionalized in 1994.
- The paper studies single retirees: never married, divorced, or widowed individuals who remain single. After sample restrictions, the authors use 3,246 individuals.
- The AHEAD contains panel information on health, nursing home stays, assets, and out-of-pocket medical expenses.
- Health is grouped into good, bad, and nursing-home states. A nursing-home state is assigned when the respondent spends a sufficiently long time in a nursing home.
- Permanent income is measured as average nonasset income, including Social Security, defined benefit pensions, veterans benefits, and annuities. SSI is excluded because it is modeled explicitly.

Interpretation: AHEAD is the core life-cycle panel. It gives the age profiles the model must match: Medicaid reciprocity, net worth, out-of-pocket medical spending, and health transitions.

3.2 MCBS data

- The Medicare Current Beneficiary Survey provides medical spending by payer: out-of-pocket, Medicaid, Medicare, and other payers.
- The paper uses the 1996-2010 MCBS waves and applies similar sample selection rules to the AHEAD sample.
- The MCBS is important because the AHEAD does not have high-quality payer-specific information on Medicaid and Medicare payments.
- The paper uses MCBS data to measure Medicaid payments and co-pay rates and to validate the model's predictions for Medicaid and total medical spending.

Interpretation: MCBS is not the main estimation data for all moments, but it is central for external validation. A model that matches Medicaid payments despite not targeting them is more credible for policy experiments.

3.3 Medicaid reciprocity, payments, and taxes

- Medicaid reciprocity is high and persistent among the poorest retirees.
- For higher-income retirees, reciprocity starts low but rises with age because very old retirees can be hit by nursing home or other catastrophic medical shocks.
- Average Medicaid benefits fall with permanent income because reciprocity falls with permanent income.
- Conditional on being a recipient, benefits rise with permanent income. Richer recipients tend to receive Medicaid only after severe, expensive events.
- Medicaid-related taxes are imputed from PSID earnings histories by allocating the general tax burden in proportion to Medicaid financing.

Interpretation: The program is redistributive on average, but it is not just a transfer to the lifelong poor. It also functions as catastrophic tail insurance for people who were not poor initially.

3.4 Medical spending and assets

- Out-of-pocket medical expenses include insurance premia, prescription drugs, hospital and nursing home costs, doctor visits, dental visits, and outpatient care.
- Median out-of-pocket spending increases late in life, especially for high permanent-income groups.
- The upper tail of out-of-pocket spending rises sharply at advanced ages, making catastrophic medical risk central to the model.
- Assets rise strongly with permanent income. High-PI retirees decumulate assets slowly, consistent with precautionary saving and bequest motives.

Interpretation: The model must explain three facts simultaneously: many poor retirees are already on Medicaid, high-PI retirees retain assets late in life, and high-PI retirees face large late-life spending risk.

4 Model and empirical specifications

4.1 Preferences and medical needs

Each period, retirees receive utility from nonmedical consumption and medical goods. A simplified representation is

$$u(c_t, m_t, \mu_t) = \frac{c_t^{1-\nu}}{1-\nu} + \mu_t \frac{m_t^{1-\omega}}{1-\omega},$$

where c_t is nonmedical consumption, m_t is medical consumption, μ_t is a medical-needs shifter, ν governs risk aversion over nonmedical consumption, and ω governs curvature over medical goods.

- Medical needs vary by age, health, nursing home status, and persistent and transitory shocks.

- Medical spending is endogenous: individuals choose medical and nonmedical consumption subject to prices, insurance, and Medicaid rules.
- The model treats medical spending as improving flow utility rather than directly changing mortality.

Interpretation: Endogenous medical spending matters because Medicaid affects not only saving but also the demand for medical care. This is the channel through which moral hazard can appear.

4.2 Insurance and Medicaid floors

The model includes two major insurance mechanisms.

- **Medicare and private insurance.** These pay a proportional share of medical spending. The individual pays a co-pay rate $q(h_t)$, which is much higher in nursing homes.
- **Medicaid.** Medicaid is means-tested and guarantees a minimum utility level. The transfer depends on financial resources and medical needs.

The categorically needy pathway applies to retirees satisfying SSI income and asset tests. The medically needy pathway applies to retirees who fail the categorical tests but have medical needs large enough to exhaust available resources.

Interpretation: The utility-floor formulation generalizes the common consumption-floor approach. A person with high medical needs requires more spending to reach the same utility floor, so Medicaid transfers naturally rise with medical needs.

4.3 Dynamic problem

At each age, a retiree observes health and medical need shocks, chooses whether to apply for Medicaid if eligible, chooses medical and nonmedical spending, saves, and then faces mortality risk. If the retiree dies, remaining assets enter a bequest utility function.

The value function can be summarized as

$$V_t(a_t, g, h_t, I, \zeta_t, \xi_t) = \max_{c_t, m_t, a_{t+1}, IM_t} \{u(c_t, m_t, \mu_t) + \beta s_t E_t[V_{t+1}] + \beta(1 - s_t)\phi(e_{t+1})\},$$

subject to the budget constraint, Medicaid eligibility rules, and the constraint that Medicaid recipients must keep assets below the applicable asset limit.

Interpretation: The problem links Medicaid to savings directly. Medicaid lowers downside consumption risk, which reduces precautionary saving, but bequest motives and late-life catastrophic risk can still sustain asset holdings.

4.4 Estimation

The paper uses a two-step approach.

1. Parameters that can be estimated outside the model are estimated first: health transitions, survival probabilities, income profiles, and co-pay rates.

2. Preference parameters, Medicaid floors, bequest parameters, and the medical-needs process are estimated by method of simulated moments.

The model matches life-cycle profiles of:

- Medicaid reciprocity by permanent income, cohort, and age;
- median assets by permanent income, cohort, and age;
- median and ninetieth percentile out-of-pocket medical spending;
- autocorrelations of medical spending.

Interpretation: The estimation forces the model to fit the distribution, not just average behavior. This matters because Medicaid is about tails: medical shocks, nursing home stays, and asset exhaustion.

5 Identification and empirical design

5.1 What identifies Medicaid generosity and bequest motives?

- Medicaid reciprocity rates discipline the Medicaid utility floors. A low floor would generate too few Medicaid recipients, especially among higher-PI retirees who enter through the medically needy route.
- A more generous Medicaid floor reduces medical expense risk and therefore reduces the need for precautionary saving.
- To match the observed high asset holdings when Medicaid is generous, the model needs a bequest motive.
- This creates an identification link: Medicaid reciprocity helps pin down the floor, and the combination of a generous floor with high assets identifies bequests.

5.2 What identifies risk aversion and medical demand?

- The slopes of asset profiles by age, cohort, and permanent income help identify patience, risk aversion, and bequests.
- The permanent-income gradient in medical spending helps identify the relative curvature parameters ν and ω .
- The median and ninetieth percentile of out-of-pocket spending identify the level and dispersion of medical need shocks.
- The autocorrelation of medical spending identifies persistent versus transitory components of medical needs.

5.3 Policy counterfactuals

The paper evaluates Medicaid policy using compensating variation. For a reform, the compensating variation Δ is the asset amount that makes a retiree indifferent between current Medicaid and the

reformed Medicaid program:

$$V_{74}(a_{74}; \text{current Medicaid}) = V_{74}(a_{74} + \Delta; \text{reformed Medicaid}).$$

- A cut in Medicaid is evaluated by the compensation needed to restore utility.
- An expansion is evaluated by the amount retirees would be willing to pay for the higher benefit.
- The paper compares these valuations to both actuarial costs and Medicaid-related tax burdens.

Interpretation: This design separates three concepts that are often confused: expected transfers, insurance value, and tax cost.

6 Tables and figures

6.1 Figure 1 and Table 1: Medicaid reciprocity and benefits

Read:

- Figure 1 shows that Medicaid reciprocity is very high for the bottom PI quintile and much lower for the top PI quintile at younger old ages.
- Reciprocity for high-PI retirees rises sharply at very old ages, consistent with medically needy entry after severe shocks.
- Table 1 shows average Medicaid benefits fall with permanent income, from \$9,080 at the bottom quintile to \$1,280 at the top quintile.
- Conditional on being a recipient, however, benefits per recipient rise with permanent income, from \$12,990 at the bottom to \$23,790 at the top.

Economic message: Medicaid has two faces: a frequently used insurance program for the poor and a rare but high-value catastrophic insurance program for richer retirees.

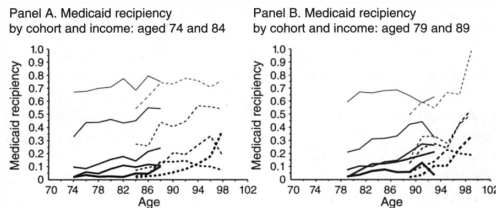


FIGURE 1. MEDICAID RECIPROCIITY RATES BY COHORT, INCOME, AND AGE

Notes: Each line represents Medicaid reciprocity rates for a cohort-PI cell, traced over the time period 1996–2010. Thicker lines refer to higher permanent income groups. Panel A: cohorts

TABLE 1—AVERAGE MEDICAID BENEFITS AND RECIPROCIITY

Permanent income quintile	Average benefit	Reciprocity rate	Average benefit per recipient
Bottom	9,080	0.70	12,990
Fourth	5,720	0.42	13,690
Third	2,850	0.16	18,350
Second	1,950	0.08	24,360
Top	1,280	0.05	23,790

Source: MCBS

(a) Figure 1: Medicaid reciprocity rates

(b) Table 1: Benefits and reciprocity

6.2 Figures 2 and 3: Out-of-pocket medical spending risk

Read:

- Figure 2 shows median out-of-pocket medical expenses by cohort, income, and age.

- Figure 3 shows the ninetieth percentile and therefore highlights tail risk.
- The upper tail of spending rises steeply at advanced ages, especially among high-PI groups.
- Medicaid protects low-PI households from out-of-pocket exposure, so their profiles are flatter.

Economic message: The model must match not only average medical spending but also the severe late-life tail risk that makes insurance valuable.

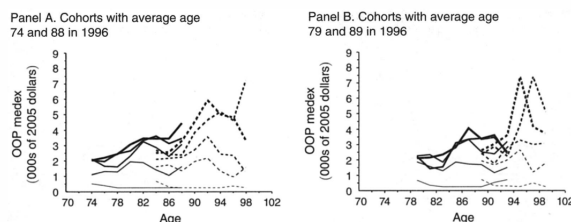


FIGURE 2. MEDIAN OUT-OF-POCKET MEDICAL EXPENSES BY COHORT, INCOME, AND AGE

Notes: Each line represents median out-of-pocket medical expenditures for a cohort-PI cell, traced over the time period 1996–2010. Thicker lines refer to higher permanent income groups. Panel A: cohorts aged 74 and 84 in 1996. Panel B: cohorts aged 79 and 89 in 1996.

(a) Figure 2: Median out-of-pocket expenses

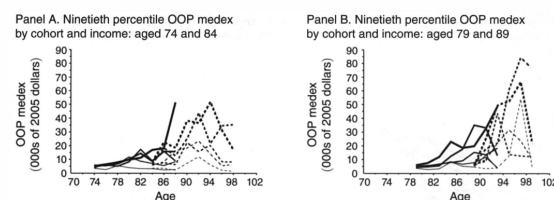


FIGURE 3. NINETIETH PERCENTILE OF OUT-OF-POCKET MEDICAL EXPENSES BY COHORT, INCOME, AND AGE

Notes: Each line represents the ninetieth percentile of out-of-pocket medical expenditures for a cohort-PI cell, traced over the time period 1996–2010. Thicker lines refer to higher permanent income groups. Panel A: cohorts aged 74 and 84 in 1996. Panel B: cohorts aged 79 and 89 in 1996.

(b) Figure 3: Ninetieth percentile out-of-pocket expenses

6.3 Figures 4 and 5: Assets and income profiles

Read:

- Figure 4 documents that assets are strongly increasing in permanent income.
- High-PI retirees hold sizable assets and do not quickly run them down even at old ages.
- Figure 5 shows model-implied income profiles by PI quintile, ranging from roughly \$5,000 per year at the bottom to above \$20,000 at the top.
- These income and asset gradients are central for distinguishing Medicaid reciprocity, precautionary saving, and bequest motives.

Economic message: The slow decumulation of assets among the elderly cannot be explained by Medicaid alone; the estimated model needs medical risk and bequest motives.

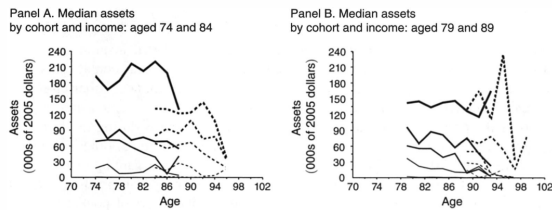
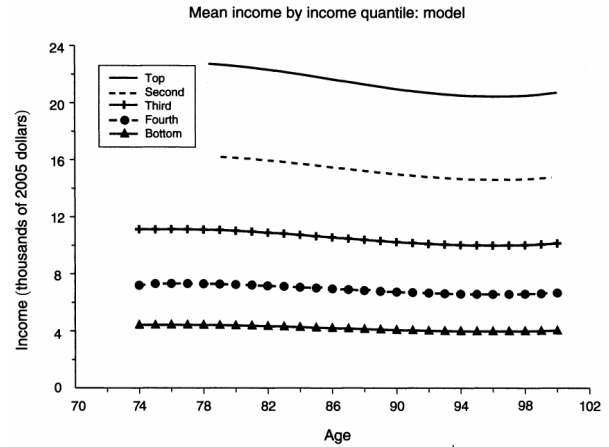


FIGURE 4. MEDIAN ASSETS BY COHORT, INCOME, AND AGE

Notes: Each line represents median assets for a cohort-PI cell, traced over the time period 1996–2010. Thicker lines refer to higher permanent income groups. Panel A: cohorts aged 74 and 84 in 1996. Panel B: cohorts aged 79 and 89 in 2006.

(a) Figure 4: Median assets



(b) Figure 5: Average income by PI quintile

6.4 Tables 2 and 3: Mortality and nursing-home risk

Read:

- Table 2 reports life expectancy at age 70 by gender, PI, and health status.
- Women, richer individuals, and healthier individuals live longer.
- Table 3 reports the probability of ever entering a nursing home.
- Nursing-home risk is high, especially for women, with roughly 46 percent of women ever entering a nursing home in the simulations.

Economic message: Longevity and nursing-home risk are the foundation of Medicaid's catastrophic value. Rich retirees have more to lose because they live longer and face late-life medical tail risk.

TABLE 2—LIFE EXPECTANCY IN YEARS

Permanent income percentile	Males			Females			All ^a
	Nursing home	Bad health	Good health	Nursing home	Bad health	Good health	
10	1.7	6.0	7.5	2.5	10.0	12.0	10.4
30	1.7	6.6	8.5	2.6	11.0	13.2	11.5
50	1.7	7.3	9.5	2.7	12.0	14.3	12.5
70	1.7	8.0	10.4	2.9	13.0	15.3	13.5
90	1.8	8.8	11.3	3.0	13.9	16.2	14.4
<i>By gender^b</i>							
Men							9.7
Women							13.5
<i>By health status^c</i>							
Bad health							10.7
Good health							14.0

Notes: Conditional on reaching age 70. Life expectancies calculated through simulations using estimated health transition and survivor functions.

^aUsing gender and health distributions for entire population.

^bUsing health and PI distributions for each gender.

^cUsing gender and PI distributions for each health status group.

(a) Table 2: Life expectancy

Permanent income percentile	Males		Females		All ^a
	Bad health	Good health	Bad health	Good health	
10	26.4	30.1	41.2	45.2	40.8
30	26.9	31.2	42.5	46.8	42.1
50	27.2	32.0	43.6	47.9	43.3
70	27.2	32.5	44.1	48.8	43.9
90	27.2	32.4	44.4	49.0	43.9
<i>By gender^b</i>					
Men					30.5
Women					46.0
<i>By health status^c</i>					
Bad health					39.9
Good health					44.9

Notes: Conditional on reaching age 70. Percentages calculated through simulations using estimated health transition and survivor functions.

^aUsing gender and health distributions for entire population.

^bUsing health and PI distributions for each gender.

^cUsing gender and PI distributions for each health status group.

(b) Table 3: Nursing-home entry risk

6.5 Table 4 and Figure 6: Estimated parameters and Medicaid model fit

Read:

- Table 4 reports the estimated preference parameters, including a high annual discount factor, curvature parameters near 3, a Medicaid utility floor of about \$4,600, and a meaningful bequest motive.
- The estimated SSI income level is close to statutory values.
- Figure 6 compares Medicaid reciprocity in the data and model.
- The model captures the broad pattern: high reciprocity among poor retirees and increasing reciprocity at very old ages among richer retirees.

Economic message: Matching Medicaid reciprocity forces Medicaid to be generous enough to matter. Once Medicaid reduces risk, the bequest motive becomes important for matching assets.

TABLE 4—ESTIMATED PREFERENCE PARAMETERS

β : discount factor	0.994 (0.013)
ν : RRA, consumption	2.83 (0.03)
ω : RRA, medical expenditures	2.99 (0.03)
$\bar{Y}$: SSI income level	\$6,670 (208)
$\underline{u}_c = \underline{u}_m$: utility floor	\$4,600 (144)
θ : bequest intensity	39.7 (2.5)
k : bequest curvature (in 000s)	13.0 (0.7)

Notes: Standard errors in parentheses. The utility floor is

(a) Table 4: Preference parameters

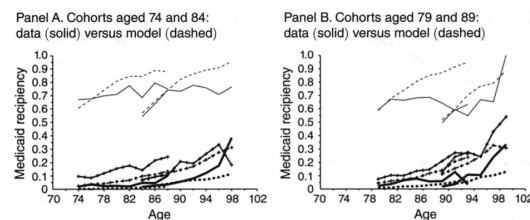


FIGURE 6. MEDICAID RECIPROCIY: DATA VERSUS MODEL

Notes: Each line represents Medicaid reciprocity for a cohort-PI cell, traced over the time period 1996–2010: data (solid lines) and model (dashed lines). For each cohort we show profiles for the bottom, third, and top PI quintiles. Thicker lines refer to higher PI groups. Panel A: cohorts aged 74 and 84 in 1996. Panel B: cohorts aged 79 and 89 in 1996.

(b) Figure 6: Medicaid reciprocity, data versus model

6.6 Figures 7 and 8: Asset model fit

Read:

- Figure 7 shows that the model matches median net worth profiles across cohorts and PI groups reasonably well.
- The model reproduces the tendency of high-PI people to hold assets late in life.
- Figure 8 compares the cross-sectional asset distribution in the model and AHEAD data.
- The model modestly underpredicts low-asset households but fits the rest of the distribution fairly closely.

Economic message: The asset fit is central because Medicaid valuation depends on how close households are to asset exhaustion and Medicaid eligibility.

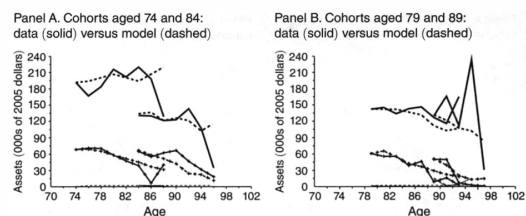


FIGURE 7. MEDIAN NET WORTH: DATA VERSUS MODEL

Notes: Each line represents median net worth for a cohort-PI cell, traced over the time period 1996–2010: data (solid lines) and model (dashed lines). For each cohort we show profiles for the bottom, third, and top PI quintiles. Thicker lines refer to higher PI groups. Panel A: cohorts aged 74 and 84 in 1996. Panel B: cohorts aged 79 and 85 in 1996.

(a) Figure 7: Median net worth, data versus model

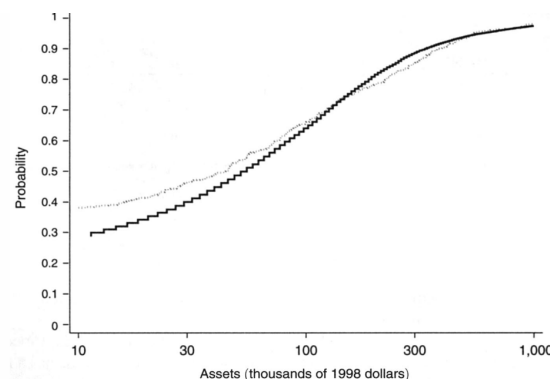


FIGURE 8. CUMULATIVE DISTRIBUTION FUNCTIONS OF ASSETS

(b) Figure 8: Asset CDF

6.7 Figures 9 and 10: Medical spending model fit

Read:

- Figure 9 shows that the model does a reasonable job matching median and ninetieth percentile out-of-pocket spending profiles.
- The model captures the flat spending profiles for lower-PI retirees and the steep late-life increase for higher-PI retirees, although it understates some high-PI tail spending in the late nineties.
- Figure 10 validates the model against cross-sectional CDFs for out-of-pocket, Medicaid, and total medical spending.
- Medicaid and total medical spending are not directly targeted in the GMM criterion, so matching these CDFs is an important external validation.

Economic message: The policy experiments rely on the model's ability to represent catastrophic medical spending. Figure 10 is therefore more than a fit check; it supports the welfare analysis.

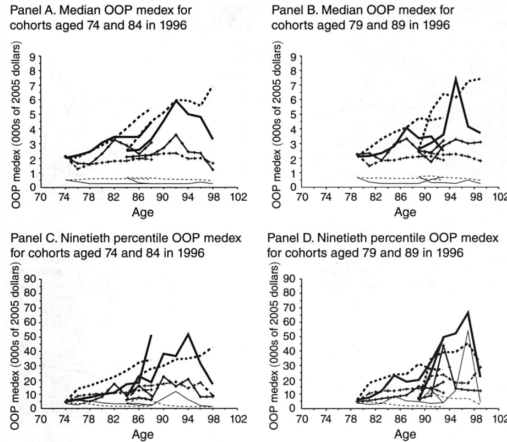


FIGURE 9. OUT-OF-POCKET MEDICAL EXPENDITURES BY COHORT AND INCOME

Notes: Each line represents the median (top panels) or ninetyeth percentile (bottom panels) of medical expenditures for a cohort-P1 cell, traced over 1996-2010: data (solid lines) and model (dashed lines). Left versus right panels:

(a) Figure 9: Out-of-pocket spending fit

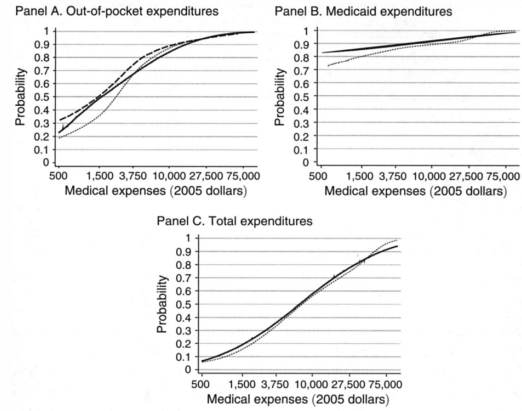


FIGURE 10. CUMULATIVE DISTRIBUTION FUNCTIONS OF MEDICAL SPENDING: DATA VERSUS MODEL

Notes: Cumulative distribution functions of medical spending: model (solid line), AHEAD data (long dashed line, panel A only) and MCBS (short dashed line). Panel A: Out-of-pocket expenditures. Panel B: Medicaid expenditures. Panel C: Total expenditures.

(b) Figure 10: Medical spending CDFs

6.8 Table 5 and Figure 11: Average spending and simulated consumption

Read:

- Table 5 compares Medicaid payments and out-of-pocket expenses in the data and model.
- The model matches average Medicaid payments fairly well, even though Medicaid payments are not directly targeted.
- Figure 11 shows model-implied nonmedical consumption for the youngest cohort.
- Nonmedical consumption falls only gradually with age, while medical spending rises sharply.

Economic message: The model separates two very different life-cycle patterns: nonmedical consumption is relatively smooth, while medical spending rises with age and risk.

TABLE 5—AVERAGE MEDICAID PAYMENTS AND OUT-OF-POCKET EXPENDITURES: DATA VERSUS MODEL

Permanent income quintile	Medicaid payments		Out-of-pocket expenses		
	MCBS data	Model	MCBS data	AHEAD data	Model
Bottom	9,080	10,070	4,050	2,550	2,210
Fourth	5,720	7,960	5,340	4,270	3,800
Third	2,850	6,000	6,470	5,050	6,330
Second	1,950	3,910	7,300	6,360	8,500
Top	1,280	2,250	8,020	7,000	10,600
Men	2,850	3,780	5,440	4,760	8,280
Women	4,410	5,980	6,470	5,230	6,420

Note: Average Medicaid payments and out-of-pocket medical expenditures (2005 dollars), model MCBS data and AHEAD data 1996-2010 for all individuals 72 and older in 1996

(a) Table 5: Medicaid and out-of-pocket expenditures

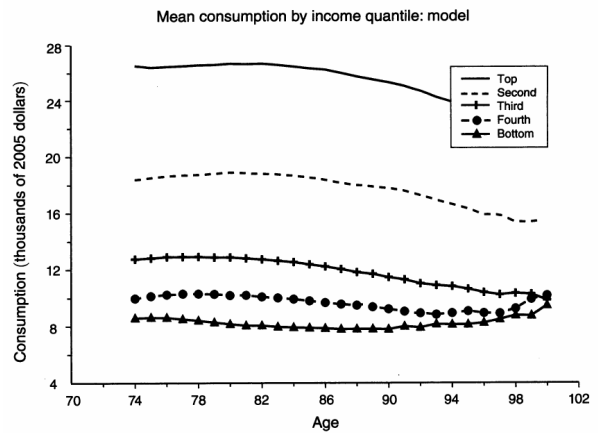


FIGURE 11. SIMULATED CONSUMPTION FOR THE COHORT AGED 74 IN 1996

(b) Figure 11: Simulated consumption

6.9 Tables 6 and 7: Lifetime spending, transfers, and taxes

Read:

- Table 6 reports present discounted values of nonmedical consumption and medical goods at age 74.
- Medical goods are large relative to nonmedical consumption, especially for low-PI retirees.
- Table 7 reports present discounted Medicaid payments and Medicaid-related taxes.
- Medicaid payments decrease with PI, but taxes rise sharply with PI. The top quintile pays about \$4.59 in taxes per dollar of expected Medicaid payments.

Economic message: Medicaid is redistributive through the tax system, but redistribution and insurance value point in different directions. Rich retirees value the insurance highly but pay much more in taxes.

TABLE 6—PRESENT DISCOUNTED VALUE OF NONMEDICAL CONSUMPTION AND CONSUMPTION OF MEDICAL GOODS AND SERVICES AT AGE 74

Permanent income quintile	Nonmedical consumption	Medical goods and services	
		Total	Out-of-pocket
Bottom	59,200	108,300	11,200
Fourth	79,700	121,100	20,400
Third	106,800	139,500	35,100
Second	163,900	178,800	55,200
Top	234,900	229,700	80,600
Men	136,000	133,900	42,700
Women	143,800	172,200	46,300
Good health	173,200	182,200	54,300
Bad health	97,500	144,000	33,000

(a) Table 6: Present discounted consumption and medical goods

Permanent income quintile	Medicaid payments	Medicaid taxes	Taxes/payments
Bottom	33,600	6,700	0.20
Fourth	29,400	8,600	0.29
Third	20,400	20,600	1.01
Second	15,100	30,300	2.00
Top	8,800	40,200	4.59
Men	8,600	28,800	3.34
Women	22,400	20,000	0.89
Good health	17,800	26,400	1.48
Bad health	23,700	12,500	0.53

Note: Present discounted value of Medicaid payments received (simulated from the model), Medicaid taxes paid (computed from the PSID), and the ratio of Medicaid taxes to Medicaid payments, all from the standpoint of age 74.

(b) Table 7: Medicaid payments and taxes

6.10 Tables 8 and 9: Medicaid cuts and expansions

Read:

- Table 8 considers a 10 percent cut in the categorically needy floor and a 10 percent cut in both Medicaid floors.
- Cutting only the categorical floor mainly affects the bottom two quintiles.
- Cutting both floors affects all groups and reveals very high insurance value for high-PI retirees, especially the top quintile.
- Table 9 considers a 10 percent increase in Medicaid generosity.
- Expansions are valued at less than cost for the bottom two quintiles, slightly above cost for middle groups, and twice cost for the top quintile, but taxes change the political interpretation.

Economic message: Current Medicaid has insurance value above its expected transfer cost, but marginal expansions are not generally valued above their cost. This is the core “approximately right size” result.

Permanent income quintile	Categorical floor down 10 percent			Both floors down 10 percent		
	Reduction in PDV of payments (1)	Compensating variation (2)	Ratio of (2)/(1) (3)	Reduction in PDV of payments (4)	Compensating variation (5)	Ratio of (5)/(4) (6)
Bottom	4,100	5,600	1.37	4,500	6,300	1.40
Fourth	2,100	2,200	1.05	4,000	5,000	1.25
Third	0.0	0.0	N/A	2,900	4,400	1.52
Second	0.0	0.0	N/A	2,200	4,100	1.86
Top	0.0	0.0	N/A	1,400	4,400	3.14
Men	300	200	0.67	1,300	1,100	0.85
Women	1,200	1,600	1.33	3,100	5,600	1.81
Good health	700	900	1.29	2,600	4,800	1.85
Bad health	1,700	2,200	1.29	3,300	5,000	1.52

Notes: Left panel: the categorically needy floor is cut by 10 percent. Right panel: both Medicaid floors are cut by 10 percent. Columns 1 and 4: decrease in the present discounted value of Medicaid payments as of age 74. Columns 2 and 5: dollar amount needed to compensate people for the Medicaid benefit cut. Columns 3 and 6: ratio of column 2 to column 1 and column 5 to column 4, respectively, which give the average compensating variation per dollar of reduced Medicaid benefits.

(a) Table 8: Cutting Medicaid

Permanent income quintile	Both floors up 10 percent		
	Payment increase (1)	Compensating variation (2)	Ratio (2)/(1) (3)
Bottom	4,700	2,600	0.55
Fourth	4,200	3,100	0.74
Third	3,100	3,600	1.16
Second	2,300	2,900	1.26
Top	1,300	2,600	2.00
Men	1,400	600	0.43
Women	3,300	3,500	1.06
Good health	2,500	3,000	1.20
Bad health	3,500	3,000	0.86

Notes: Column 1: increase in the present discounted value of Medicaid payments at age 74. Column 2: dollar amount people would be willing to pay to receive the higher Medicaid benefits. Column 3 is the ratio of column 2 to column 1, which show the average compensating

(b) Table 9: Increasing Medicaid

6.11 Tables 10 and 11: Moral hazard and selection into Medicaid

Read:

- Table 10 studies what happens at age 85 when Medicaid payments are reduced by 10 percent.
- The Medicaid cut mechanically reduces transfers and eligibility, while behavioral responses in nonmedical spending and out-of-pocket spending are modest.
- This suggests mechanical insurance effects are larger than moral hazard effects in the policy experiment.
- Table 11 compares Medicaid recipients with nonrecipients in simulations.
- Medicaid recipients have much lower income, much higher total medical spending, and much higher medical need shocks.

Economic message: Medicaid recipients are selected on severe medical needs. This supports the interpretation of Medicaid as last-resort insurance against catastrophic late-life events.

TABLE 10—THE EFFECTS OF DECREASING MEDICAID PAYMENTS BY 10 PERCENT, AGE 85

Permanent income	Consumption	Medical expenditures		Medicaid reciprocity (percent)
		Total	OOP	
Baseline				
Bottom	7,890	22,110	1,430	89.1
Fourth	9,810	21,280	2,860	42.9
Third	12,380	22,710	5,310	9.9
Second	18,520	25,630	8,210	3.7
Top	26,330	32,790	11,860	1.3
Medicaid floors decreased 10 percent				
Bottom	7,600	20,320	1,580	88.6
Fourth	9,600	19,870	3,010	41.0
Third	12,240	21,580	5,420	8.9
Second	18,420	24,980	8,260	3.3
Top	26,210	32,360	11,850	1.2
Difference				
Bottom	-290	-1,790	150	-0.5
Fourth	-210	-1,410	150	-2.0
Third	-140	-1,130	110	-1.0
Second	-100	-650	50	-0.4
Top	-120	-430	-10	-0.1
Percentage differences				
Bottom	-3.7	-8.1	10.5	-0.6
Fourth	-2.1	-6.6	5.2	-4.5
Third	-1.1	-5.0	2.1	-9.6
Second	-0.5	-2.5	0.6	-10.6
Top	-0.5	-1.3	-0.1	-7.1

(a) Table 10: Effects of decreasing Medicaid at age 85

TABLE 11—COMPARISON OF MEDICAID AND NON-MEDICAID RECIPIENTS: SIMULATIONS

	All retirees	Medicaid recipients	Non-recipients
Mean net income	16,270	6,240	19,020
Mean medical spending			
Total	28,930	61,820	19,920
Out-of-pocket	6,680	2,940	7,700
Medicaid	5,680	26,440	0
Medical needs shifter (μ)			
Mean	1,270	5,150	210
Percent in top 50 percent of distribution	50.0	67.9	45.1
Percent in top 10 percent of distribution	10.0	26.0	5.6
Percent in top 1 percent of distribution	1.0	4.0	0.1
Percent of population	100.0	19.7	80.3

(b) Table 11: Medicaid versus non-Medicaid recipients

7 Short summary

- The paper studies Medicaid insurance for currently single retirees.
- It combines AHEAD, MCBS, and PSID data to measure assets, medical spending, Medicaid transfers, Medicaid reciprocity, and taxes.
- It estimates a life-cycle model with endogenous medical spending, saving, bequests, mortality risk, health risk, nursing-home risk, and two Medicaid eligibility pathways.
- Medicaid reciprocity is high among the poor and rises at advanced ages among higher-income retirees.
- Average Medicaid payments fall with permanent income, but payments conditional on reciprocity are larger for higher-income recipients.
- The model matches the main life-cycle profiles and validates well against Medicaid payment and total medical spending data.
- Current Medicaid insurance is valued above its actuarial cost, especially because of catastrophic medical risk.
- Marginal expansions are generally valued below their actuarial cost, implying that the current program is roughly the right size for current single retirees.
- Medicaid is redistributive through taxes, so high-income retirees value the insurance but pay much more to finance it.

8 Conclusion

8.1 Core conclusion

De Nardi, French, and Jones show that Medicaid in old age is both redistributive insurance for low-income retirees and catastrophic insurance for higher-income retirees. The model implies that retirees value existing Medicaid insurance at more than the expected transfer cost. However, most retirees would not value a marginal expansion at its cost. This leads to the conclusion that the current Medicaid program is approximately the right size for the population studied.

8.2 Economic intuition

The poor use Medicaid often because they satisfy categorical income and asset tests. The rich use Medicaid rarely, but when they do, they are typically very old and have experienced extremely expensive medical conditions. Because the utility value of insurance is highest in states with catastrophic losses, rich retirees can value Medicaid transfers very highly even though their expected transfers are small.

8.3 Incremental contribution revisited

The paper advances the literature by integrating realistic Medicaid rules, endogenous medical spending, nursing home risk, rich heterogeneity, and lifetime valuation in one structural framework. Its main conceptual insight is that Medicaid cannot be evaluated only by average transfers. The value of insurance depends on rare states of the world, especially very old age combined with high medical needs.

8.4 Policy implications

The analysis suggests caution about both large cuts and simple expansions. Cuts can impose welfare losses larger than fiscal savings because they remove insurance against catastrophic risk. Expansions may be valued less than their fiscal cost, especially when they mostly raise benefits in states where existing insurance already provides substantial protection. Tax financing also matters: the groups that value Medicaid expansions most may also pay the most to finance them.

8.5 Limitations

The model focuses on single retirees rather than couples or the full life cycle. It abstracts from endogenous long-term care insurance purchases, even though Medicaid may crowd out private long-term care insurance. It also takes much of the tax system as an allocation exercise rather than modeling the full distortionary cost of taxation. Finally, as in any structural model, the counterfactual results depend on the assumed preferences, medical spending technology, and Medicaid rules.

8.6 Takeaway

The key takeaway is that Medicaid for the elderly is not merely a means-tested transfer program. It is a state-contingent insurance system against the most severe late-life medical shocks. Its redistributive costs and insurance benefits differ sharply across the permanent-income distribution, and evaluating the program requires modeling both sides together.